

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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LAB EXP: 1

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Course Code/ Name: DSA0601 / Data Handling and Visualization

Set 1: Monthly Sales Data

Dataset: Monthly Sales

```
# =====  
# SET 1: MONTHLY SALES DATA  
# =====  
  
# Load required libraries  
library(ggplot2)  
library(dplyr)  
library(plotly)  
library(shiny)  
library(tm)  
library(wordcloud)  
library(RColorBrewer)  
  
# -----  
# DATASET: MONTHLY SALES  
# -----  
  
sales_data <- data.frame(  
  Month = c("January", "February", "March", "April", "May"),  
  Sales = c(15000, 18000, 22000, 20000, 23000)  
)  
  
# Convert Month into ordered factor  
sales_data$Month <- factor(  
  sales_data$Month,  
  levels = c("January", "February", "March", "April", "May")  
)  
  
print(sales_data)
```

OUTPUT:-

```
> print(sales_data)
```

Month Sales

1 January 15000

2 February 18000

3 March 22000

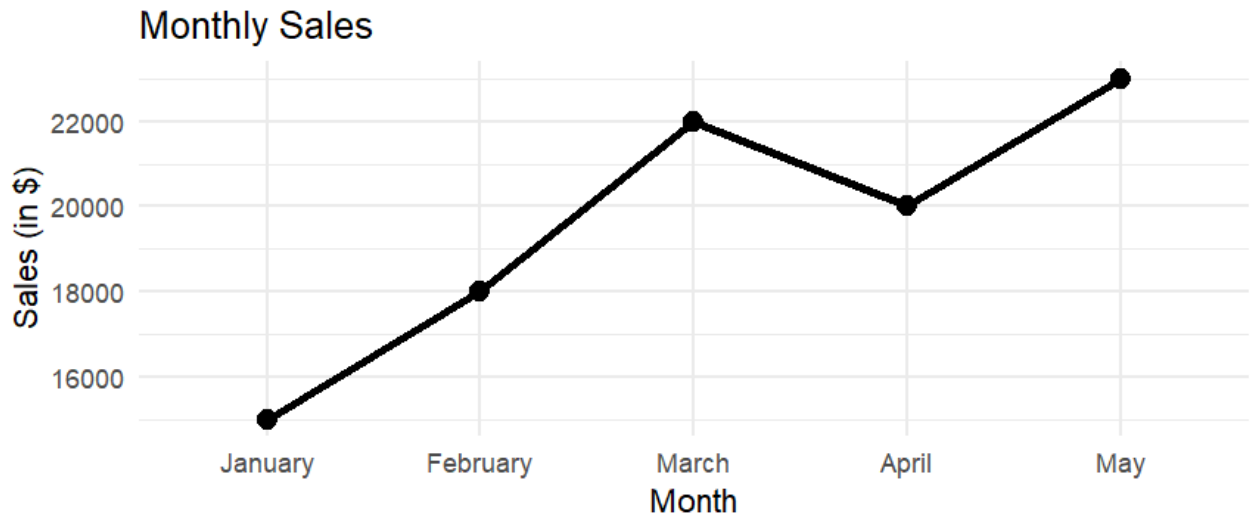
4 April 20000

5 May 23000

1. Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

```
# =====  
# SET 1 - QUESTION 1  
# Create a LINE CHART to visualize monthly sales  
# =====  
  
line_chart <- ggplot(sales_data,  
                     aes(x = Month, y = Sales, group = 1)) +  
  geom_line(linewidth = 1.2) +  
  geom_point(size = 3) +  
  labs(  
    title = "Monthly Sales",  
    x = "Month",  
    y = "Sales (in $)"  
  ) +  
  theme_minimal()  
|  
print(line_chart)
```

OUTPUT:-



2. Generate a bar chart to display the top-selling products for the year. Label the chart elements.

```
# =====
# SET 1 - QUESTION 2
# BAR CHART - TOP SELLING PRODUCTS FOR THE YEAR
# =====

# The original question does not provide product-wise data,
# so sample product data is created.

product_data <- data.frame(
  Product = c(
    "Laptop",
    "Mobile Phone",
    "Headphones",
    "Smart Watch",
    "Tablet"
  ),
  Sales = c(
    85000,
    78000,
    45000,
    38000,
    32000
  )
)

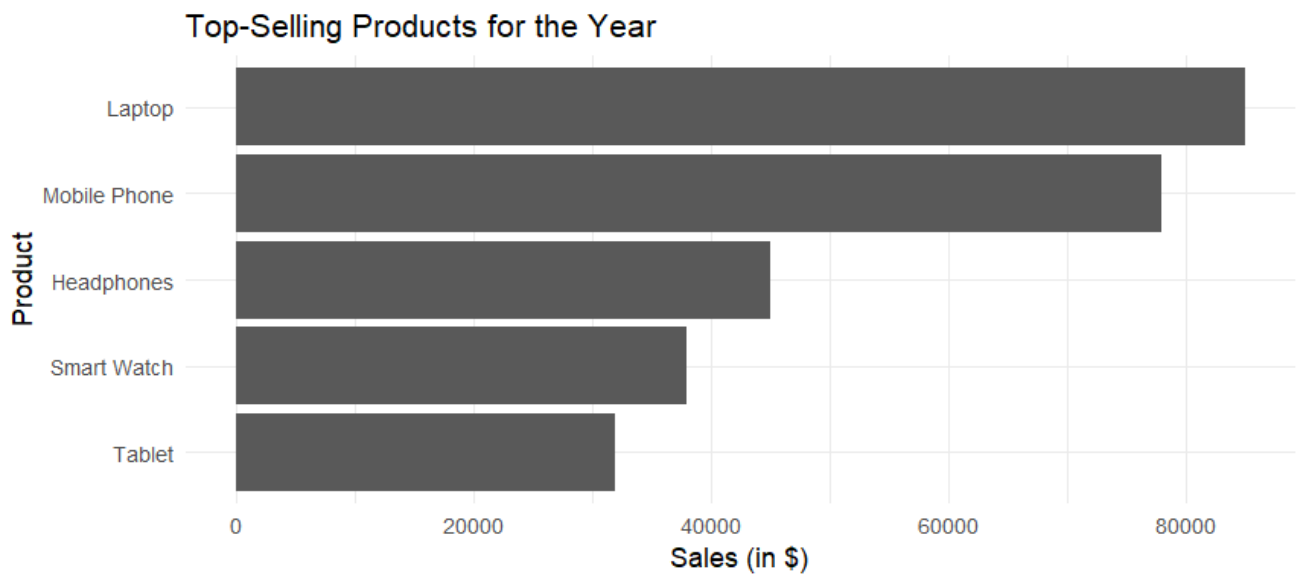
print(product_data)

# Arrange products from highest to lowest
product_data <- product_data %>%
  arrange(desc(Sales))

# Bar chart
bar_chart <- ggplot(
  product_data,
  aes(
    x = reorder(Product, Sales),
    y = Sales
  )
) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Top-Selling Products for the Year",
    x = "Product",
    y = "Sales (in $)"
  ) +
  theme_minimal()

print(bar_chart)
```

OUTPUT:-



3. Develop a scatter plot to explore the relationship between advertising budget and monthly sales.

```

# =====
# SET 1 - QUESTION 3
# SCATTER PLOT
# Relationship between Advertising Budget and Monthly Sales
# =====

# Sample advertising budget data
advertising_data <- data.frame(
  Month = c("January", "February", "March", "April", "May"),
  Advertising_Budget = c(
    3000,
    4000,
    5000,
    4500,
    6000
  ),
  Sales = c(
    15000,
    18000,
    22000,
    20000,
    23000
  )
)

print(advertising_data)

# Scatter plot
scatter_plot <- ggplot(
  advertising_data,
  aes(
    x = Advertising_Budget,
    y = Sales
  )
) +
  geom_point(size = 4) +
  geom_smooth(
    method = "lm",
    se = FALSE
  ) +
  labs(
    title = "Advertising Budget vs Monthly Sales",
    x = "Advertising Budget ($)",
    y = "Monthly Sales ($)"
  ) +
  theme_minimal()

print(scatter_plot)

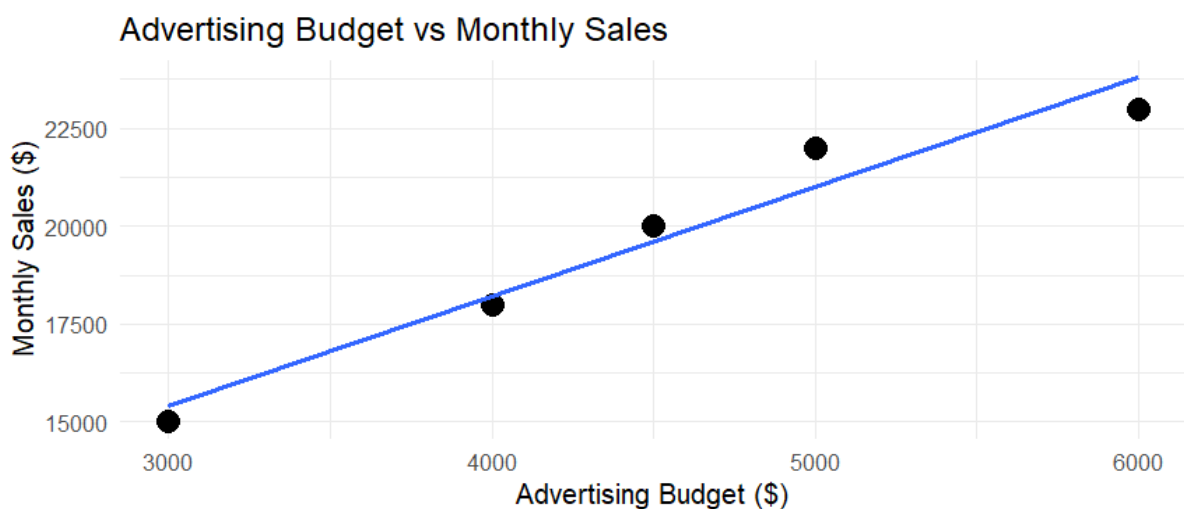
# Correlation
correlation <- cor(
  advertising_data$Advertising_Budget,
  advertising_data$Sales
)

cat(
  "\nCorrelation between Advertising Budget and Sales:",
  correlation,
  "\n"
)

# Interpretation
if (correlation > 0) {
  cat(
    "Interpretation: There is a positive relationship between ",
    "advertising budget and monthly sales.\n"
  )
}

```

OUTPUT:-



4. Build an interactive dashboard combining the line chart and bar chart to allow users to explore sales data interactively.

```
# =====  
# SET 1 - QUESTION 4  
# INTERACTIVE DASHBOARD  
# Combining Line Chart and Bar Chart  
# =====  
  
# UI  
ui <- fluidPage(  
  titlePanel("Monthly Sales Interactive Dashboard"),  
  sidebarLayout(  
    sidebarPanel(  
      h4("Dashboard Controls"),  
      selectInput(  
        inputId = "chart_type",  
        label = "Select Chart:",  
        choices = c(  
          "Monthly Sales",  
          "Top-Selling Products"  
        ),  
        selected = "Monthly Sales"  
      ),  
    ),  
    mainPanel(  
      plotlyOutput(  
        outputId = "sales_plot",  
        height = "600px"  
      )  
    )  
  )  
)  
  
# Server  
server <- function(input, output) {  
  output$sales_plot <- renderPlotly({  
    if (input$chart_type == "Monthly Sales") {  
      p <- ggplot(  
        sales_data,  
        aes(  
          x = Month,  
          y = Sales,  
          group = 1,  
          text = paste(  
            "Month:", Month,  
            "<br>Sales: $", Sales  
          )  
        )  
      ) +  
        geom_line(linewidth = 1.2) +  
        geom_point(size = 4) +  
        labs(  
          title = "Monthly Sales",  
          x = "Month",  
          y = "Sales ($)"  
        ) +  
        theme_minimal()  
    }
```

OUTPUT:-

